

Relative Reporting

When the headline is a ratio and the benefit is missing

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Fault 1 of six · healthy-skepticism.com · September 2026

The fault

The fault is relative reporting. A benefit is stated as a percentage change, and the percentage sounds far larger than the number of people actually helped. The example this course opens with: a statin advertisement says "Cuts the risk of heart attack by 36%." The sentence is built to impress, and it does. A drug that removes a third of your chance of a heart attack sounds like a drug everyone should be taking. That is what the reader takes away, and it is not what the trial found.

The claim isn't false. It is overblown. The 36% is a real number from a real study, reported in the form that makes the drug sound as good as it possibly can, while the number a patient would need to judge the benefit is left out. A reader who takes the headline at face value pictures a benefit many times larger than the one that was measured. This paper shows what the 36% really means, why the claim was reported this way, and how a healthy skeptic reads a number like it.

The example: a statin advertisement

The advertisement that opened the Healthy Skepticism course led with a 36% reduction in heart attacks. The number is not fabricated. It is a real result from a real trial, in which about ten thousand people with high blood pressure and average cholesterol were followed for a little over three years. Expressed the way this course expresses everything, whole people out of a stated number of people, the trial looks like this. Of 1,000 people like those in the trial taking the placebo, about 30 had a heart attack. Of 1,000 taking the drug, about 19. Eleven fewer per thousand.



Of 1,000 people like those in the trial: 30 heart attacks without the drug, 19 with it.

Divide the eleven by the thirty and you get the 36%. That is where the headline number comes from: the drop in the rate, measured against the rate. The eleven is the benefit; the 36% is the ratio; and the ratio is what was printed. Both are true. Only one of them lets a patient decide.

The honest sentence: Of 1,000 people like those in the trial, about 30 had a heart attack without the drug and about 19 with it. Eleven avoided one. About 91 people had to take the drug for one of them to benefit.

That last figure has a name, the number needed to treat: 1,000 divided by 11, about 91. It is the most useful number in the report and it almost never makes the headline.

ASCOT-LLA: Sever PS et al. Prevention of coronary and stroke events with atorvastatin in hypertensive patients. Lancet 2003;361:1149–1158. Event counts rounded to whole people per 1,000 for teaching.

Why the baseline matters

The same relative number describes benefits a thousandfold apart. A 50% reduction of a risk of 2 in 10,000 removes one case in ten thousand. The same 50% applied to a risk of 2 in 10 removes one case in ten. The headline is identical; the number of people helped differs by three orders of magnitude. A relative figure without its baseline is not yet a fact; it is half of one.

The fault runs the other way too. Harms are often reported in absolute terms when they are small (“one extra case per ten thousand”) and benefits in relative terms when they are small, because each framing makes its number look the way the writer needs it to look. The test is not whether a percentage or a count was used; it is whether the reader can recover both.

Why it happens

Nobody has to lie. The trial is real, the arithmetic is correct, and the ratio is a legitimate statistic. The fault is a choice of which number to show, and the choice follows an incentive: the most impressive-sounding number makes the headline, and the number that would let you judge stays in the fine print. Drug advertising, press releases, and news coverage are all written by people who benefit when the reader is impressed, and a ratio computed against a small baseline is reliably impressive. The baseline isn’t hidden outright; it is in the trial paper. The advertisement simply separates the ratio from the number it is a ratio of and prints only the ratio.

On the relative-versus-absolute framing effect in physicians and patients: Gigerenzer G et al. Helping doctors and patients make sense of health statistics. Psychol Sci Public Interest 2007;8:53–96.

What to ask

- 36% of what? A percentage with no stated baseline is not yet a fact.
- What was the risk without the treatment? This is the number the ratio was computed from.
- Out of how many people, how many actually benefited? Ask for it as whole people out of a thousand.
- If a thousand people like me took this, how many would be helped, and how many would have had the problem anyway?

If a claim survives all four, it is probably reported fairly. If any of the answers isn’t available, be suspicious of the claim, not of the medicine.